# Explain It Like You Built It — Background Jobs

**Name: Sibgha Mursaleen**
**Date: 16/08/2026**

In my project, I moved the slow AI normalization process into a background job. Before this, the API had to wait for the LLM to finish before returning the result. With the new **/jobs** endpoint, when a user sends a job title, the API creates a unique **job_id**, saves the job with a **queued status**, and immediately returns **202 Accepted** with that ID. This means the user doesn't have to wait for the LLM response.

After that, FastAPI runs the **process_job_task** in the background. The job changes from queued to running, and the worker calls my existing LLM client to normalize the job title. When the LLM returns a result, the worker stores it in my in-memory **jobs_db** and changes the status to completed. If something goes wrong, the job can be retried or eventually marked as failed.

The user can then use **GET /jobs/{job_id}** with the job ID they received earlier. The API looks up that specific job in **jobs_db** and returns its current status, result, error, and number of attempts. So basically, instead of making the user wait for a slow AI operation, my system accepts the request quickly, processes it in the background, and lets the user check the result later using the job ID.